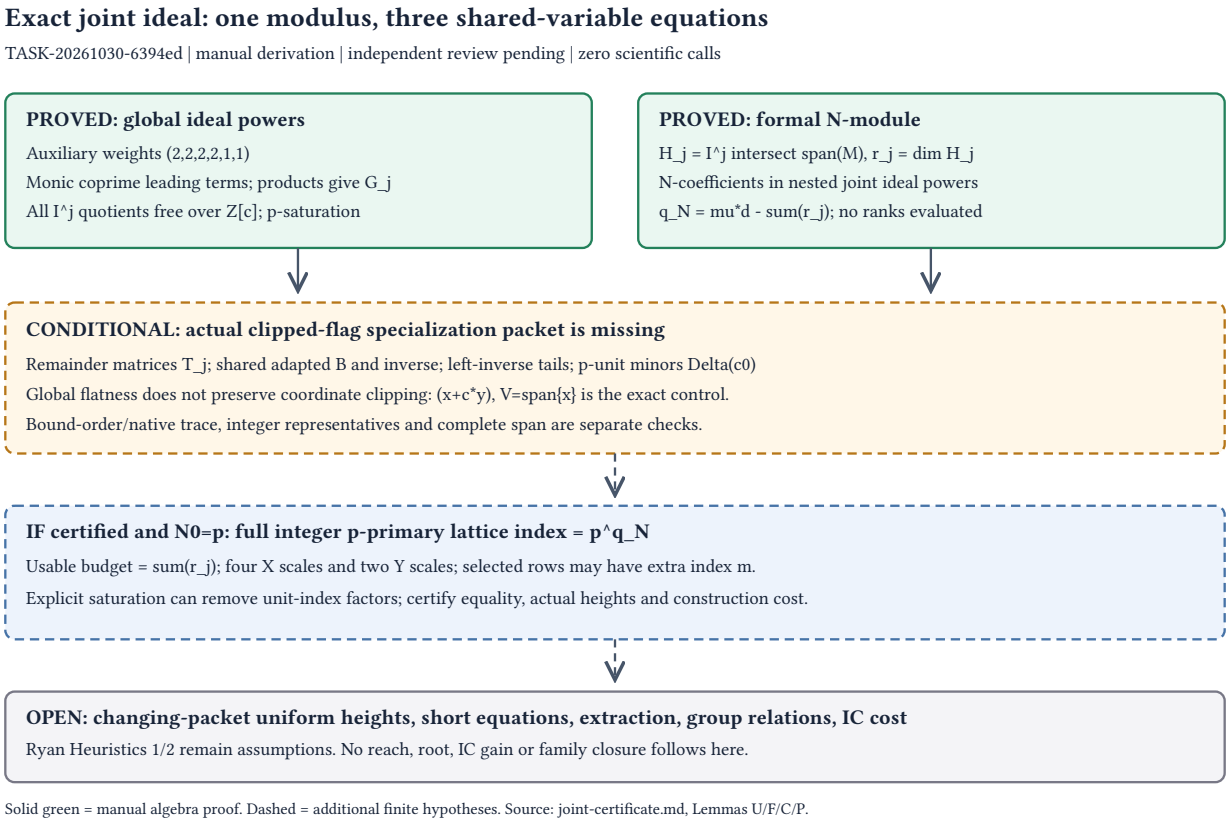
Joint ideal layers and specialization

A conditional finite certificate · 30 October 2026

TASK-20261030-6394ed · GOAL-PFDR-fd09e0 · BATCH-ee839d

Manual derivation; independent review pending. Zero scientific calls.

The exact three-equation, six-variable Semaev ideal has an explicit global monic ideal-power control. A fixed finite domain additionally needs a clipped-intersection certificate before its formal modulus exponents can be transferred to a full integer lattice index. No ranks, actual cell, numerical reach, root or IC gain were evaluated.



The full proof is retained in **joint-certificate.md**. This PDF is a concise overview; the separately retained certificate-schema.yaml, theorem-matrix.yaml, proof-search-map.yaml and source-manifest.json specify exact conditions and provenance.

The input is I=(F1,F2,F3), Jμ=(N,I)^μ, with formal N independent of all coefficient parameters, J∞=0 and x=(z1,z2,z3,z4,h1,h2). The protocol fixes F1=S(c1+z1,c2+z2,d1+h1), F2=S(c3+z3,c4+z4,d2+h2), F3=S(d1+h1,d2+h2,q). There is one modulus. The four leaf bounds X and two internal bounds Y remain distinct.

Source and evidence status: primary source reading and manual algebra only. The report is a derivation, not an executed or independently reviewed certificate. Root owns publication and the subsequent distinct review sessions.

What is proved, and what is conditional

Global structure for the exact equations. Over A0=Z[a,b,q,c1,c2,c3,c4,d1,d2], use auxiliary x-weights (2,2,2,2,1,1). The unique top terms of F1,F2,F3 are respectively $z1^2z2^2$, $z3^2z4^2$, $h1^2h2^2$, each coefficient 1. The child top weight is 8 versus at most 7 for competing terms; the target top weight is 4 versus at most 3. Centers lower weights.

These three monomials are pairwise coprime. Introduce independent coefficient variables t1,t2,t3. The monic product criterion makes F_i−t_i a Gröbner basis over A0[t][x]. Its standard x-monomials W form a free A0[t]-basis of the quotient, which is A0[x] with t_i acting by F_i. Each x-monomial uniquely factors as a product of the three leading monomials times w in W. The resulting F^β w have distinct monic leading terms. Therefore the products F^β with total exponent j are a monic Gröbner basis for I^j, and A0[x]/I^j is free over A0. This holds after arbitrary coefficient specialization. It proves global p-saturation of the ideal powers; no regular-sequence assumption is made.

The auxiliary order does not reproduce the bound-induced native order. It certifies ideal membership and global flatness. It does not enumerate the actual domain M or determine a clipped rank.

Exact formal-modulus module. Fix an explicit finite monomial set M, d=|M|, and V=span(M). Define the joint flag H_j=I^j∩V and r_j=dim over Q(c). In an expansion g=Σ N^a g_a, ideal membership is exactly g_a in I raised to max(μ−a,0). An adapted flag basis v_i therefore gives supported module basis N^λ_i v_i, with

q_N = Σ λ_i = μd − Σ from j=1 to μ of r_j.

This is a formal N-order. A coefficient-field unit can have large integer size or nonzero specialized p-valuation. The identity alone is not an integer determinant formula.

Finite clipped certificate. Reduce every m in M by the proved monic product basis of I^j. Let T_j contain the resulting standard remainder columns. Supply a common invertible matrix B over A0[1/Δ] with inverse, and identities T_j B=(0,C_j), L_j C_j=identity. These prove that each kernel is exactly the corresponding prefix of B and remains so under every allowed base change. Membership-only vectors or generic row rank are insufficient.

For each fixed finite μ,M, such a nonzero Δ and finite identities exist after localization. The actual T_j, ranks, B, left inverses and Δ have not been produced for the protocol cells. The quantifiers are “for each fixed packet there exists a finite localization, then every good specialization.” No one Δ, uniform packet size, exceptional-fiber frequency or changing-multiplicity efficiency is inferred.

Prime layers, integer span and height

Conditional bridge to the full integer lattice. Put N0=p prime and specialize parameters to integers with p not dividing the clipped certificate's Δ. Work over O=Z_(p). Global power saturation proves ((p,I)^t:p)=(p,I)^(t−1). The clipped base-change identities then prove L_t=H_t+p L_(t−1), where L_t=(p,I)^t∩O^M. Its exact p-layers have dimensions r_(μ−a). The complete supported integer lattice contains p^μ Z^M, so its index has only the prime p and is exactly p^q_N. Its weighted determinant is

p^q_N × product over e in M of X^(e1+…+e4) Y^(e5+e6).

Its usable determinant-level budget is Σ r_j. At μ=2, the ten product generators give q_N=2d−r1−r2. No rank value was evaluated. Suppressing F2,F3 reproduces the archived principal accounting K=t·d−q; the one-polynomial Newton-polytope ceilings do not thereby transfer to the joint ideal.

Selected integer rows can have an additional index m in the complete lattice, multiplying their determinant by m. Rational rank does not show equality; p-local equality only shows p does not divide m. If raw rows have certified full p-local span, their prime-to-p saturation equals the full p-primary lattice. This can remove unit-index losses; actual saturation/congruence-preimage construction, output heights and costs still need evidence. Denominator clearing is a possible raw-row loss, not an unavoidable loss for every implementation.

Exact controls. Generic (N,cx)=(N,x), but at c=p,N=p,x=1 the normalized row x evaluates to 1, while both original generators evaluate to p. Clearing the denominator gives valid cx, with coefficient size p. A matrix with rows (1,1) and (1,1+c) has denominator 1 and rational rank two at c=p, but residue rank one. The monic ideal (x+c y), clipped to span(x), has generic intersection zero and intersection span(x) at c=0. At c=p it fails residue clipping despite a free global quotient. These controls separate units, rank minors and clipped base change. Repeating a relation leaves the ideal unchanged.

Integer heights. For H bounding 1 and all nine integer parameters, the exact affine inputs satisfy coefficient L1 norm at most 28(H+1)^4. If a retained symbolic row has cleared numerator P_i=d_i g_i with an integral ideal witness, its specialized L1 norm is at most norm1(P_i) H^degree_c(P_i) max(1,|N0|)^degree_N(P_i). Charge these degrees/norms, denominator sizes, transformation coefficients and any unremoved row index.

Alternatively, invert d_i modulo |N0|^μ and center each coefficient of the resulting row. Its absolute coefficients are at most |N0|^μ/2; subtracting multiples of N0^μ preserves membership. Actual leading terms and integer span must be checked again. Products of k certified base rows have multiplicity μk and product L1 norm bounds; coefficient reduction must respect N0^(μk). Fixed-packet bounds with H≤p^C give polynomial bit lengths in k and log p, but a changing μ,M packet needs new uniform bounds.

Source boundaries and the exact next obligation

Ryan primary, actually read. Keegan Ryan, "Solving Multivariate Coppersmith Problems with Known Moduli," ePrint 2024/1577, pinned 52-page PDF, archived primary URL in source-manifest.json; SHA256 begins 15dd6ae6b939c51f and is recorded in full there. Read: §2.5 / Heuristic 1 p6; §3 pp7–8; §4.3 pp11–12; §§6.1–6.2 pp19–20; §6.4 / Heuristic 2 pp21–22; Lemma 7 p23; Algorithm 3 p24; Appendix C pp48–51. Full PDF/text stay external, not in Git.

Section 6.1 requires coefficient denominators to be invertible modulo N raised to the precomputation multiplicity. Section 4′s optimal integer lattice is defined by integer span; it is not automatic proof that coefficient-field source rows specialize to a complete integer basis. Heuristic 1 assumes zero-dimensionality of the recovered integer ideal. Heuristic 2 assumes the stated polynomial dimension/determinant functions for suitable sets. Lemma 7 retains these assumptions, shared-root/suitability conditions, proportional specialization, strict bound and its growing-log-p condition. Its polynomial-time statement **excludes final root recovery**. Appendix C's fixed-precomputation coefficient bounds do not certify an arbitrary changing base packet.

Pinned source reading. cuso commit fb2747d2105a8e48499bf10a3a717cf80c14d079 was inspected read-only: symbolic problem, convert, ideal, optimal, asymptotic, shift_set_properties, limit and utils. Its returned-row denominator list is narrower than this design's full witness/minor/span packet. The native model.hpp bytes matched published owner commit 4f27d0cd0a50b94d4f990a593321cf2951015ce9; no unpublished integration was adopted. Historical assigned reviews and upstream S3 §§1–2 / S4–S5 §§7b–7c were read with their original scope and qualifications.

No knowledge MCP is exposed. Exact repository rg retrieval covered the assigned batch, historical joint audit, lead/problem records and generated indexes. This is a bounded retrieval scope, not an exhaustive novelty claim. The auxiliary-order and clipped (x+c y) directions were suggested by the parent after the initial flag derivation and checked manually by this producer. No independent-review status is claimed.

JOINT-FINITE-PACKET-6394ed: supply the actual fixed M, monic remainders T_j, ranks, shared B/inverse, tail left inverses and Δ; prove native row membership and both supported-span directions. At a concrete prime, check units, the actual bound-order trace, integer representatives, full integer index/saturation and heights. This would qualify a finite algebraic interface only.

JOINT-UNIFORM-HEIGHT-6394ed: restrict to a fixed finite base packet or prove explicit uniform degree/norm/size/good-locus and construction-cost bounds for the claimed varying μ,k,M and coefficients. No random-parameter success frequency is derived.

Actual short integer equations, tractable extraction, F_p y-lifts and signs, subgroup/target equality, random-target yield and independent relations, and matched fully charged IC/rho costs remain open. The existing problem is joint-smallroot-reach-baseline-and-oracle-unverified.md. No global problem, canonical graph or frozen record was edited.

Scientific/algorithm calls, trials, ideal/rank/domain computations, workers and provisioned hosts: **0**. All 24 old cells remain unrun; Q4 is terminal no-retry; prior independence exit1 is unwaived. No state promotion or family closure. Contribution: conditional proof interface; no quantitative advantage. Full nine-artifact source/provenance and rendering inspection are recorded beside this PDF.